React Lifecycle Methods in Class Components

React lifecycle methods allow you to hook into different stages of a component's life: from initialization to rendering, updating, and eventually unmounting. Below is a breakdown of the various lifecycle methods used in class components.

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1. Mounting Phase

These methods are called when the component is being created and inserted into the DOM.

1.1. constructor(props)

- Purpose: Used to initialize the component's state and bind event handlers.

Syntax:
js
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```
constructor(props) {
  super(props);
  this.state = { count: 0 }; // Initializing state
  this.handleClick = this.handleClick.bind(this); // Binding methods
}
```

-
- When to Use: Typically used for setting initial state and binding methods before the component is mounted.

1.2. static getDerivedStateFromProps(props, state)

- Purpose: Synchronizes the component's state with changes in props before each render.

Syntax:
js

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```
static getDerivedStateFromProps(nextProps, nextState) {
  if (nextProps.someValue !== nextState.someValue) {
    return { someValue: nextProps.someValue }; // Update state based on props
  }
  return null; // No update to state
}
```

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- When to Use: Useful when the state needs to be updated in response to prop changes before rendering.

1.3. render()

- Purpose: Renders the component's UI. It returns JSX, which will be converted into DOM elements.

Syntax:
js
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```
render() {
  return <div>{this.state.someState}</div>;
}
```

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- When to Use: The core method for rendering the UI. It is mandatory for class components.

1.4. componentDidMount()

- Purpose: Called once the component has been rendered and inserted into the DOM. It's used for performing side-effects like data fetching, subscribing to events, or manipulating the DOM.

Syntax:
js
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```
componentDidMount() {
  fetchData().then(data => this.setState({ data }));
}
```

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- When to Use: Ideal for AJAX calls, setting up subscriptions, or directly interacting with the DOM (e.g., initializing third-party libraries).

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2. Updating Phase

These methods are invoked when a component's state or props change.

2.1. static getDerivedStateFromProps(props, state) (again)

- Purpose: Called before every render to sync the state with props.
- Syntax: Same as the Mounting phase.

2.2. shouldComponentUpdate(nextProps, nextState)

- Purpose: Decides whether the component should re-render in response to state or prop changes. It is a performance optimization tool.

Syntax:
js
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```
shouldComponentUpdate(nextProps, nextState) {
  return nextState.someValue !== this.state.someValue; // Only update if state changes
}
```

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- When to Use: For performance optimization by preventing unnecessary re-renders when props or state do not change.

2.3. render()

- Purpose: Returns the JSX to render the UI. This method will be called every time there is a state or prop change.
- Syntax: Same as the Mounting phase.

2.4. getSnapshotBeforeUpdate(prevProps, prevState)

- Purpose: Called before the DOM is updated, this method lets you capture information (such as scroll position) before the changes are applied to the DOM.

Syntax:
js
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```
getSnapshotBeforeUpdate(prevProps, prevState) {
  return this.someRef.scrollTop; // Capture scroll position before update
}
```

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- When to Use: Useful for capturing some information (like scroll position) before the update happens in the DOM.

2.5. componentDidUpdate(prevProps, prevState, snapshot)

- Purpose: Called after the component is updated. It can be used for side-effects such as making network requests or updating the DOM based on previous props or state.

Syntax:
js
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```
componentDidUpdate(prevProps, prevState, snapshot) {
  if (this.state.someState !== prevState.someState) {
    console.log('State has changed!');
  }
}
```

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- When to Use: Ideal for responding to prop or state changes and performing additional actions, such as network requests or DOM manipulation.

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3. Unmounting Phase

These methods are called when a component is being removed from the DOM.

3.1. componentWillUnmount()

- Purpose: This is where you perform any necessary cleanup, such as clearing timers, canceling network requests, or unsubscribing from events.

Syntax:
js

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```
componentWillUnmount() {
  clearInterval(this.timer); // Clean up resources before unmounting
}
```

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- When to Use: Use for cleanup tasks that prevent memory leaks, like canceling subscriptions or clearing timers.

Summary of React Lifecycle Methods

Phase	Method	Purpose
Mounting	constructor(props)	Initialize state and bind methods.
	static getDerivedStateFromProps()	Sync state with props before each render.
	render()	Returns JSX to render the UI.
	componentDidMount()	Fetch data, set up subscriptions, etc.
Updating	static getDerivedStateFromProps()	Sync state with props during updates.
	shouldComponentUpdate()	Optimize performance by preventing re-renders.
	render()	Re-renders component with updated state/props.
	getSnapshotBeforeUpdate()	Capture data (e.g., scroll position) before DOM update.
	componentDidUpdate()	Perform side-effects after component update.
Unmounting	componentWillUnmount()	Clean up resources before the component unmounts.